

Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II (MAL1020)

Problem Sheet IV
Academic Year 2025–2026, Semester II

Q.1 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined by

$$T(x, y, z) = (x + y, y + z).$$

- (a) Find the matrix representation of T with respect to the standard bases.
- (b) Find the kernel and the range of T .
- (c) Determine whether T is an onto linear transformation.

Q.2 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$$T(X) = AX,$$

where

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

- (a) Find the rank and nullity of the linear transformation T .
- (b) Is T one one and onto linear transformation?

Q.3 Let

$$W = \text{span}\{(1, 0, 2), (2, 1, 3)\} \subset \mathbb{R}^3.$$

Find the orthogonal complement $W^\perp$.

Q.4 Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \end{bmatrix},$$

and define

$$W = \text{Null}(A) \subset \mathbb{R}^3.$$

- (a) Find a basis for W .
- (b) Find the orthogonal complement $W^\perp$.

- (c) Verify that every vector in $\mathbb{R}^3$ can be written as a sum of a vector in W and a vector in $W^\perp$.

Q.5 Let $P_n(\mathbb{R})$ denote the vector space of all real polynomials of degree at most n . Define a linear transformation

$$T : P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$$

by

$$T(f(x)) = xf(x) + f'(x).$$

Determine whether T is one one and onto transformation?

Q.6 Let V be the vector space of all real 3×3 matrices, and let A be the diagonal matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Calculate the determinant of the linear transformation T on V defined by

$$T(X) = \frac{1}{2}(AX + XA).$$

Q.7

- (a) Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation such that $T(1, 0) = (1, 4)$ and $T(1, 1) = (2, 5)$. Calculate the value of $T(2, 3)$ and determine if T is one-to-one.
- (b) Prove that there exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfying $T(1, 1) = (1, 0, 2)$ and $T(2, 3) = (1, -1, 4)$. Find the value of $T(8, 11)$.
- (c) Determine whether there exists a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(1, 0, 3) = (1, 1)$ and $T(-2, 0, -6) = (2, 1)$.

Q.8

- (a) Consider a linear transformation $T : P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by:

$$T(f(x)) = \begin{pmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{pmatrix}$$

Construct a matrix representation w.r.t. Standard basis. Find the basis of the Kernel and the Image of T . Also verify the Rank-Nullity theorem.

- (b) $T : M_{2 \times 3}(F) \rightarrow M_{2 \times 2}(F)$ defined by:

$$T \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{pmatrix}.$$

Do the same as the above problem (Matrix representation, Kernel, Image, Rank-Nullity).